Personalized AI Plutus Tutor - Project Documentation

Project Overview

The Personalized AI Plutus Tutor is an intelligent, adaptive learning system designed to tailor Plutus smart contract education based on a user's expertise level and personality type. The system fine-tunes learning materials dynamically, ensuring an optimized and inclusive learning experience for diverse learners, including logical thinkers, hands-on practitioners, and neurodiverse individuals.

Key Features

- **Adaptive Learning Pathways:** Users select their expertise level (Beginner, Intermediate, Advanced) and personality type (e.g., Hands-on, Visual Learner, Autistic-Friendly), and the tutor generates a personalized learning plan.
- **Fine-Tuned AI Model:** The tutor is built on a fine-tuned LLM trained with multiple variations of Plutus learning content to generate explanations that match different learning preferences.
- **Dynamic Content Generation:** Instead of static pre-written materials, the AI dynamically generates explanations, code snippets, and examples suited to the user's profile.
- **Retrieval-Augmented Generation (RAG):** The system integrates structured Plutus documentation to enhance the accuracy and contextual relevance of AI-generated responses.
- **Interactive Coding & Assessments:** Users engage with quizzes, exercises, and coding challenges tailored to their cognitive style and learning pace.

Technical Architecture

1. Data Preparation & Training

- **Dataset Structure:**
  - JSON dataset containing Plutus course modules categorized by expertise level and personality type.
  - Examples of tailored content include step-by-step instructions for neurodiverse users, visual explanations for visual learners, and interactive coding prompts for hands-on learners.

- **Fine-Tuning Approach:**
  - Model: LLaMA 7B, Mistral, or OpenAI fine-tuned with LoRA (Low-Rank Adaptation) for efficient specialization.
  - Training: Data is preprocessed, tokenized, and used to train the AI model with distinct response styles.

2. AI Tutor Implementation

- **LLM Agent:**
  - Receives user input (expertise + personality type + topic).
  - Retrieves relevant learning material and generates responses dynamically.
- **Retrieval-Augmented Generation (RAG):**
  - Enhances response accuracy by referencing stored documentation and structured learning materials.
- **Backend API:**
  - Deploys the AI model via FastAPI for real-time interactions.
  - Provides endpoints for retrieving customized learning content.

3. Deployment & User Interaction

- **Frontend Web Interface:**
  - Users select their learning preferences via an intuitive UI.
  - AI tutor dynamically generates content, quizzes, and exercises.
- **Feedback Mechanism:**
  - Users provide feedback on content relevance, improving future responses through reinforcement learning.
- **Integration Options:**
  - API access for external applications.
  - Potential integration into e-learning platforms.

Project Goals & Benefits

Educational Impact

- **Improves Plutus Adoption:** Enables a more inclusive and engaging way to learn Plutus smart contract development.
- **Accommodates Diverse Learners:** Supports different cognitive styles and learning needs.
- **Scalable AI-Powered Learning:** Reduces the need for human tutors while maintaining personalization.

Development Roadmap

1. **Phase 1:** Data Collection & Model Training

   - Prepare structured datasets for AI fine-tuning.
   - Train a base LLM with multiple learning variations.
2. **Phase 2:** Prototype Development
   - Deploy a working prototype with a functional UI.
   - Integrate API-based learning content retrieval.
3. **Phase 3:** Testing & Refinement
   - Gather user feedback for model improvements.
   - Enhance interactivity with better exercises & assessments.
4. **Phase 4:** Deployment & Scaling
   - Deploy on cloud infrastructure for public access.

FUTURE SCOPE

- Expand content coverage beyond Plutus basics.

Our approach not only enhances learning outcomes but also fosters broader adoption of Plutus and Cardano blockchain technologies.

For further inquiries or collaboration opportunities, please reach out to the project team
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